

MidTerm Q

There are 5 **essay questions** – 论述题.

没有选择题!

Time: 11:30~13:00

1. Describe **three populations** of users with **special needs**. For each of these populations, suggest **three** ways current interfaces **could be improved to** better serve them(O).

Users with Hearing Impairments 损伤

Challenges:

- May struggle with video content **lacking captions**.
- Cannot access audio cues, spoken instructions, or alerts.

Improvement:

- **Provide accurate captions and transcripts** for all audio and video content.
- **Use visual alerts** (e.g., flashing icons, banners) instead of sound notifications.
- **Enable sign language avatars or overlays**

Users with Visual Impairments (e.g. Color-blind User)

Challenges:

- Difficulty distinguishing certain color combinations (e.g., red-green, blue-yellow).
- Low contrast between elements can hinder readability and interaction.

Interface Improvements:

- **Avoid using color as the only means of communication:** Always pair color with text labels, icons, or shapes.
 - Example: Instead of showing a red error message alone, include a warning icon and the word “Error.”
- **Use color-blind safe palettes** Choose combinations that are distinguishable for most types of color blindness (e.g., blue-orange, purple-yellow).
 - Avoid problematic pairs like red-green or green-brown.

- **Offer color mode toggles or simulation tools** Provide a "color-blind friendly mode" in settings.

Users with Cognitive Disabilities (e.g., dyslexia 阅读障碍, memory challenges)

Challenges:

- May struggle with complex layouts, jargon, or multi-step processes.
- Need more time and clarity to process information.

Interface Improvements:

- **Use plain language and consistent terminology** to reduce cognitive load.
- **Break tasks into small, manageable steps** with clear progress indicators.
- **Offer customizable interfaces** (e.g., simplified mode, reading assistance, icon-based navigation).

Older Adults

Challenges:

- **Declining vision:** Difficulty reading small text or low-contrast elements.
- **Cognitive load:** Slower information processing, memory limitations, and unfamiliarity with digital conventions.

Improvements:

- **Use large, readable text and high contrast**
- **Simplify task flows and reduce steps:** Break complex processes into clear, linear steps.
- **Design large, clearly labeled buttons:** Avoid relying on icons alone — pair with text.

Users with Motor Disabilities (运动障碍用户)

Challenge:

- Inability to **precisely operate** a mouse, touchscreen, or keyboard

Improvements:

- Support **voice control**, or **switch-based input**. 开关控制
- Provide **large buttons** and **sufficiently sized touch targets**.
- Avoid **time-limited interactions** (e.g., countdown-based verification codes).

2. Suppose you need to design a system for users in both the **United States** and **Japan**. Present a list of **cultural differences** that you should be aware of so that a **successful design** can be made (O).

<https://blog.btrax.com/american-japanese-ux-design/>

- Color Usage:

In Japanese UI design, **red** is correlated to **boldness or positivity**

while in the US, red correlates to **emphasis or error**.

- Hierarchy of information(Information Density) :

Americans have **short reading attention spans**. => America utilizes **white space** in an attempt to **separate sections** of information from one another.

U.S. users prefer **minimalist, clean** layouts with clear hierarchy and **whitespace**.

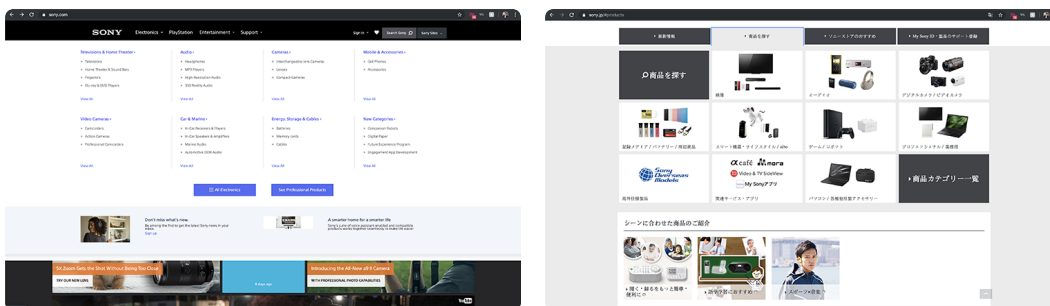
Japanese people seek information by **reading text**, => Japanese websites have more content and typography. => **Sectioned** framework.

While **Japanese users** are comfortable with **dense** information, **multiple** columns, and layered menus

- Navigation:

In Japan, The search goes from a **general search to a more specific** search on each page.

While the US, **laying out all the categories** and listing out all the departments under their respective categories first.



Error Handling and Feedback

- **U.S. users** expect direct, actionable error messages.
- **Japanese users** may prefer indirect, polite phrasing and less confrontational tone

Help-Seeking Behavior

- **U.S. users** are more likely to explore and experiment.
- **Japanese users** tend to read instructions first and avoid trial-and-error

3. Why is **recognition** better than **recall** in **minimizing the memory load** of the user? How do **UX designers** minimize the need for **recall** in UX Design?

<https://www.interaction-design.org/literature/topics/recognition-vs-recall?srltid=AfmBOooOM7jWOsQt62ix41JqfXtcg2zGPEjd9ZR5aKTo5XUImSwzl18>

Recognition and recall describe **how users retrieve information**.

- **Recognition** is the ability to recognize something you **have seen before**.
 - 看到后认出
- Recall is the ability to **remember** something **without being prompted**.
 - 听到描述后努力回忆

情境	Recognition（识别）	Recall（回忆）
登录界面	显示用户名列表供用户选择	要求用户手动输入用户名
菜单导航	显示图标和标签供用户点击	要求用户记住命令并输入（如 CLI）
密码恢复	提示安全问题供用户选择答案	要求用户记住并输入旧密码

Answer:

Why:

Recognition requires less cognitive effort: Users can identify familiar elements when they see them, rather than having to retrieve information from memory.

Recall requires more effort and can be more prone to errors: When users rely on their memory to find what they are looking for, they are more likely to become frustrated and abandon the task altogether.

How:

Use familiar icons and labels: Use standard icons and labels that users can quickly recognize.

For example, a magnifying glass icon for search or a home icon for the homepage can help users find what they are looking for more easily.

Provide visual cues: Visual cues such as color, contrast, and typography can help guide users to important information. Use different colors or font sizes for headings, subheadings, and body text to help users scan content more easily.

Follow common patterns: Design interfaces that follow common patterns to reduce the need for recall.

For example, users expect to find a logo in the top-left corner of a website.

Group related information together: This can help users find what they are looking for without the need to remember where it is located.

For example, grouping all account settings under one tab or menu item can make navigation easier for users.

4. Describe at least **three different types of expert review methods**. (Ch5)

- **Heuristic evaluation**
 - **Guidelines review**
 - **Consistency inspection**
 - **Cognitive walkthrough**
 - **Formal usability inspection**
-

Heuristic Evaluation (启发式评估)

专家根据一组通用的**可用性原则**检查界面是否存在问题。

- 快速、低成本
- 适合早期设计阶段

优点：覆盖面广，适合发现常见问题

缺点：**依赖专家经验**，可能忽略特定用户群体问题

Guidelines Review (设计指南审查)

专家将界面与**官方或行业标准**进行对照检查。【很多设计标准（如 WCAG、Apple HIG、Material Design）并不是“非黑即白”的规则，而是**带有模糊性和上下文依赖**。】

优点：确保界面符合技术标准和法规

缺点：可能忽略用户实际行为或情境差异

Consistency Inspection (一致性检查)

检查界面内部是否在术语、布局、交互方式上保持一致。

Cognitive Walkthrough (认知走查)

专家模拟新用户完成具体任务的过程，分析每一步是否容易理解和执行。

- 关注“第一次使用者”的学习路径

Formal Usability Inspection (正式可用性审查)

由多个专家组成评审小组，按照结构化流程评估界面可用性。

优点：系统性强，适合正式报告

缺点：成本高、流程复杂

【Expert review/ Usability testing 是什么？有什么用？只做一个可以吗？顺序颠倒可以吗？】

5. Give two reasons why expert reviews are useful. Also give two limitations of expert reviews. (Ch5)

Reasons (O):

a. Identify Design Flaws Early

Experts can spot usability issues, inconsistencies, and violations of design principles **before the product reaches users.**

This helps teams **save time and cost** by **avoiding rework** after deployment.

b. Leverage Domain Knowledge and Standards

Experts apply **established heuristics, guidelines, and best practices** to evaluate interfaces systematically.

Their feedback is grounded in **prior experience and research**, making it more reliable than casual user impressions.

Limitations:

The dangers with expert reviews are that the experts may **not** have an adequate understanding of the task domain or user communities.

Even experienced expert reviewers have great **difficulty** knowing how typical users, especially first-time users, will really behave

An **expert review** is a method of evaluating a product, system, or application by having one or more subject matter **experts use their knowledge to identify potential problems from a user's perspective**

专家在进行界面评估或可用性分析时，**核心职责是从用户角度出发**，但他们也必须**理解并权衡投资人和利益相关者的目标**

“Heuristic” 启发式原则：一种基于经验的评估标准，用于判断界面设计是否合理、易用、符合用户心理模型。

“First-time user” 是指第一次使用某个系统、软件或界面的人。

6. What is usability testing? (O)

Usability testing is a method of **evaluating a product** like a website or app **by observing real users** as they **complete tasks** to identify problems and opportunities for improvement.

The goal is to make the product more user-friendly and to ensure it meets user needs by **gathering direct feedback** on the user experience.

This is often done through a combination of **observing** participants, **collecting** both qualitative (user comments) and quantitative (task success rates) data, and then analyzing the results to inform design decisions.

7. “Usability testing is better than expert reviews.” Do you agree? (O)

落脚点在于阶段+需求+钱

	Usability Testing	Expert Reviews
Perspective	Real users performing real tasks	Experts applying heuristics and experience
Strengths	Reveals actual user behavior and pain points	Identifies design rule violations and consistency issues
Cost & Speed	Slower, more expensive	Faster, cheaper
Depth of Insight	Rich behavioral data, emotional reactions	Systematic, rule-based diagnosis
Best For	Validating usability with target audience	Early-stage design critique or budget-limited projects

When Usability Testing Is Better?

- **Need real user feedback** on task completion, confusion points, or emotional reactions.
- **Testing a live or high-fidelity prototype** and need to validate assumptions.
- **Targeting specific user groups** (e.g., elderly, children, professionals) whose behavior may differ from expert expectations.

When Expert Reviews Are Better?

- **In early design stages** and want fast, low-cost feedback.
- **Need to catch heuristic violations** like inconsistent layouts
- **Under time or budget constraints** and can't recruit users.

8. (O) Q3: What is think aloud protocol? What is the **difference** between concurrent think aloud and retrospective think aloud? (Ch5) 【一个case，分析用哪个好一点】

Think-aloud protocol is a technique used in usability testing and cognitive research where users are asked to *say out loud what they are thinking* as they perform tasks.

	Concurrent Think-Aloud	Retrospective Think-Aloud
Timing	During task	After task
Data freshness	Real-time	Memory-based
Task interference	Possible	None
Common use	Early-stage usability testing	Post-task analysis or interviews

9. What is gulf of execution? What is gulf of evaluation? Give an example of each.(Ch3)

- **Gulf of execution** 执行鸿沟: Mismatch between the **user's** intentions and the allowable actions 用户想做某事，但界面不支持或不清楚怎么做。
 - 例如：用户想保存文档，但找不到“保存”按钮。
- **Gulf of evaluation** 评估鸿沟: Mismatch between the **system's** representation and the user's expectations 系统的反馈方式与用户的预期不一致。
- 给 Case 判断，并解释原因。

用户预期	系统反馈	结果
红色表示错误	系统用绿色显示“错误已处理”	用户以为操作成功
点击“保存”后无提示	页面静默刷新	用户不确定是否保存成功

10. Describe three principles of direct manipulation. (Ch7)

【解释DM是什么，一个interface，判断是否符合DM】

- **Continuous** representations of the objects and actions of interest with meaningful visual metaphors 隐喻.

用连续可视化的方式呈现用户关注的对象和操作，并通过有意义的视觉隐喻来增强理解

- **Physical actions** or **presses of labeled buttons**, instead of complex syntax.

强调“可操作性”和“可识别性”，用户不需要记住命令，只需点击明确标识的按钮。

- **Rapid, incremental, reversible** actions whose effects on the objects of interest are **visible immediately**.

11. (O) Compare command language interfaces to direct manipulation interfaces with respect to compactness, speed of performance and learnability. (Ch7)

	Command Language Interface (CLI)	Direct Manipulation Interface (DM)
Compactness 用最少的表达单位（如字符、命令、界面元素）完成复杂任务的能力– 表达效率	High — 少量字符表达复杂操作 e.g., <code>rm file.txt</code>) Allow users to express complex operations using minimal symbolic input.	Low — 每个操作都需单独点击或拖动 Rely on graphical elements like buttons and icons. Each action typically requires a separate visual component and physical interaction (e.g., clicking, dragging), which consumes more screen space and interaction steps.
Speed of Performance	Fast for experts — Once mastered, commands can be executed rapidly	Moderate — Actions like dragging or clicking are slower than typing for experts
Learnability	Low — Requires memorization of syntax and commands	High — Visual metaphors and immediate feedback make it intuitive for beginners

12. (O) What is meant by **interface metaphor**? (Ch7) 【是什么+意义】

【给出一个metaphor, 问好/不好】

An interface metaphor is a design technique that uses **familiar concepts from the real world** to make computer interfaces intuitive.

For example:

- The file and folder analogy for the **file system** of an operating system
- The trash can icon represents "delete"
- The "shopping cart" for e-commerce

- **Whiteboard metaphor:** Collaborative drawing apps

Users can immediately understand how to interact with the system using their existing knowledge.

13. (O) What is the relationship between **interface metaphor** and **direct manipulation**? (Ch7)

Interface metaphors provide the conceptual **foundation** for direct manipulation by making virtual actions feel familiar and intuitive,

While **direct manipulation operationalizes** those metaphors through visible, reversible, and physical interactions.

Direct Manipulation: Interaction Style

- Direct manipulation refers to interfaces where users interact with visible objects using **physical actions** (e.g., dragging, clicking, resizing).
- Key attributes:
 - **Continuous visual representation** of objects
 - **Physical actions** replace complex commands
 - **Immediate feedback** and **reversible operations**

This style **relies on metaphors** to make interactions **feel natural**. For example, dragging a file to the trash bin is meaningful only because the metaphor makes 'delete' feel like 'throwing away.'

【举例子：good/bad metaphors】

GOOD:

Desktop metaphor	File systems, icons, trash bin
Shopping cart metaphor	E-commerce platforms
Magnifying glass icon for search	Search functions
Trash bin icon for delete	File deletion

BAD:

Potato for internet	Arbitrary, no shared meaning
Book metaphor for hypertext	Misleading for non-linear navigation

14. What are the differences between **Augmented Reality** and **Virtual Reality**? (Ch7)

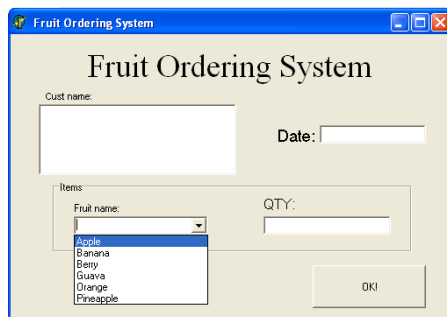
- **Virtual reality** breaks the physical limitations of space and allow users to act as though they were somewhere else
 - **Augmented reality** shows the real world with an overlay of additional overlay; Augmented reality is an important variant. Enables users to see the real world with an overlay of additional interaction.
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Related to 8 Golden Rule:

15. Give a brief **explanation** of the Eight Golden Rules of Interface Design. State an example you have seen on a device, computer interface or web site that **violates those rules**.

16. Name four of the **eight Golden Rules**, and provide one example of each.

17. Describe **three design flaws** with the interface below. Name **two of** the Eight Golden Rules that have been violated by this interface. Give a brief explanation of how they were violated



Date 注意format (mm/dd/yy) => Date picker 解决

【给一个新的design，需要去理解，要带铅笔和尺子】

试卷给出一个差的案例，需要描述出存在的问题，并设计一个好的案例，自己画上去。